

Continuous Random Variables Exercises

Mathematics for AI

Question 1. Expectation of the uniform distribution on $[a, b]$

Let $X \sim \text{Uniform}[a, b]$ with $a < b$. Calculate $\mathbb{E}[X]$.

If X is a continuous random variable that is uniform on $[a, b]$ then its density function is:

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

The expectation $\mathbb{E}[X]$ is then:

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} u f_X(u) \, du \\ &= \int_a^b u \cdot \frac{1}{b-a} \, du && (f_X = 0 \text{ outside of } [a, b]) \\ &= \frac{1}{b-a} \left[\frac{u^2}{2} \right]_a^b \\ &= \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} \\ &= \frac{(b-a)(b+a)}{2(b-a)} && (\text{factorising the difference of two squares}) \\ &= \frac{a+b}{2}. \end{aligned}$$

$$\text{So } \mathbb{E}[X] = \frac{a+b}{2}.$$

Question 2. Distribution function of the exponential distribution

Let $X \sim \text{Exponential}(\lambda)$ with $\lambda > 0$. Calculate the distribution function $F_X(x)$.

The exponential distribution has the following probability density function $f_X(x)$:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0. \end{cases}$$

The distribution function for a continuous random variable is: $F_X(x) = \int_{-\infty}^x f_X(u) \, du$.

There are 2 scenarios to consider:

Scenario 1 where $x < 0$:

$$F_X(x) = \int_{-\infty}^x 0 \, du = 0.$$

Scenario 2 where $x \geq 0$:

$$\begin{aligned} F_X(x) &= \int_{-\infty}^0 0 \, du + \int_0^x \lambda e^{-\lambda u} \, du \\ &= \left[-\frac{\lambda}{\lambda} e^{-\lambda u} \right]_0^x \\ &= \left[-e^{-\lambda u} \right]_0^x \\ &= -e^{-\lambda x} - (-e^0) \\ &= 1 - e^{-\lambda x}. \end{aligned}$$

So the distribution function for the exponential distribution is:

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0. \end{cases}$$